

Abhimanyu Hans

✉ ahans1@cs.umd.edu
🌐 [ahans30.github.io](https://github.com/ahans30)
🔗 [ahans30](#)

Research Interests

Post Training Large Language Models, Security & Privacy

Education

- 2022 – 2027 **Ph.D. and Masters**, Computer Sciences, University of Maryland, College Park
◦ Advised by Prof. Tom Goldstein
- 2014 – 2017 **Bachelors (Hons.)**, Mathematics, University of Delhi

Publications

(* indicates joint first authorship)

- ICML 2024 **Spotting LLMs With Binoculars: Zero-Shot Detection of Machine-Generated Text**
Abhimanyu Hans*, Avi Schwarzschild*, Valeriia Cherepanova, Hamid Kazemi, Aniruddha Saha, Micah Goldblum, Jonas Geiping, Tom Goldstein
41st International Conference on Machine Learning [Paper] [Code] [Demo]
- NeurIPS 2024 **Be like a Goldfish, Don't Memorize! Mitigating Memorization in Generative LLMs**
Abhimanyu Hans, Yuxin Wen, Neel Jain, John Kirchenbauer, Hamid Kazemi, Prajwal Singhanian, Siddharth Singh, Gowthami Somepalli, Jonas Geiping, Abhinav Bhatele, Tom Goldstein
38th Annual Conference on Neural Information Processing Systems [Paper] [Code]
- SC 2024 **Democratizing AI: Open-source Scalable LLM Training on GPU-based Supercomputers**
Siddharth Singh, Prajwal Singhanian, Aditya Ranjan, John Kirchenbauer, Jonas Geiping, Yuxin Wen, Neel Jain, Abhimanyu Hans, Manli Shu, Aditya Tomar, Tom Goldstein, Abhinav Bhatele
International Conference for HPC, Networking, Storage and Analysis [Paper]
- Preprint **Has My System Prompt Been Used? Large Language Model Prompt Membership Inference**
Roman Levin*, Valeriia Cherepanova*, Abhimanyu Hans*, Avi Schwarzschild, Tom Goldstein
[Paper]
- Preprint 2026 **Multi-Token Prediction via Self-Distillation**
John Kirchenbauer, Abhimanyu Hans, Brian Bartoldson, Micah Goldblum, Ashwinee Panda, Tom Goldstein [Paper]
- ICLR Workshop **Scalable Pretraining of Retrieval Models**
Khalid Saifullah, John Kirchenbauer, Rifaa Qadri, Abhimanyu Hans, Neel Jain, Yuxin Wen, Siddharth Singh, Brian R. Bartoldson, Bhavya Kailkhura, Abhinav Bhatele, Tom Goldstein

Work Experience

- Summer 2026 **Research Intern, Google Research**
Working on Contextual Integrity compliant training recipe under multi-agent interaction setup.
- Summer 2024 **Research Intern, Pangram Labs**
Developed AI-generated voice and text detection systems at Pangram Labs, improving model robustness and evaluation across diverse domains and generation models.
- 2021 – 2022 **Senior Machine Learning Engineer, PayPal India**
Built an automated model monitoring framework with interactive dashboards across 100+ model use cases, using metrics (AUC, Accuracy, KS, PSI) and LIME to improve performance and interpretability.

2020 – 2021 **Machine Learning Engineer**, *HDFC Bank*

Developed an anomalous transaction detection system (AutoEncoder, Isolation Forest) achieving 25% recall at 0.1% FPR, and built a Flask-based CRUD app with PostgreSQL to streamline model management.

2018 – 2020 **Data Scientist**, *Think360.AI*

Built deep tree models to predict construction equipment repossession values and automated a BI pipeline (Tableau, R, Bash) delivering dashboards and reports to 5000+ users.

Selected Awards, Research in News and Services

- **Golfish Loss in Atlantic Magazine** [\[link\]](#)
- **Binoculars on IEEE Spectrum** [\[link\]](#)
- **Binoculars on Hacker News** [\[link\]](#)
- **Binoculars on Business Insider** [\[link\]](#)
- **Binoculars on Freethink** [\[link\]](#)
- **Invention of the Year** for Binoculars Patent in Maryland Innovate 2024 [\[link\]](#)
- **Best Performer Award**, IIIT Bangalore 2023
- **Reviewer** for ICML, ICLR, & NeurIPS